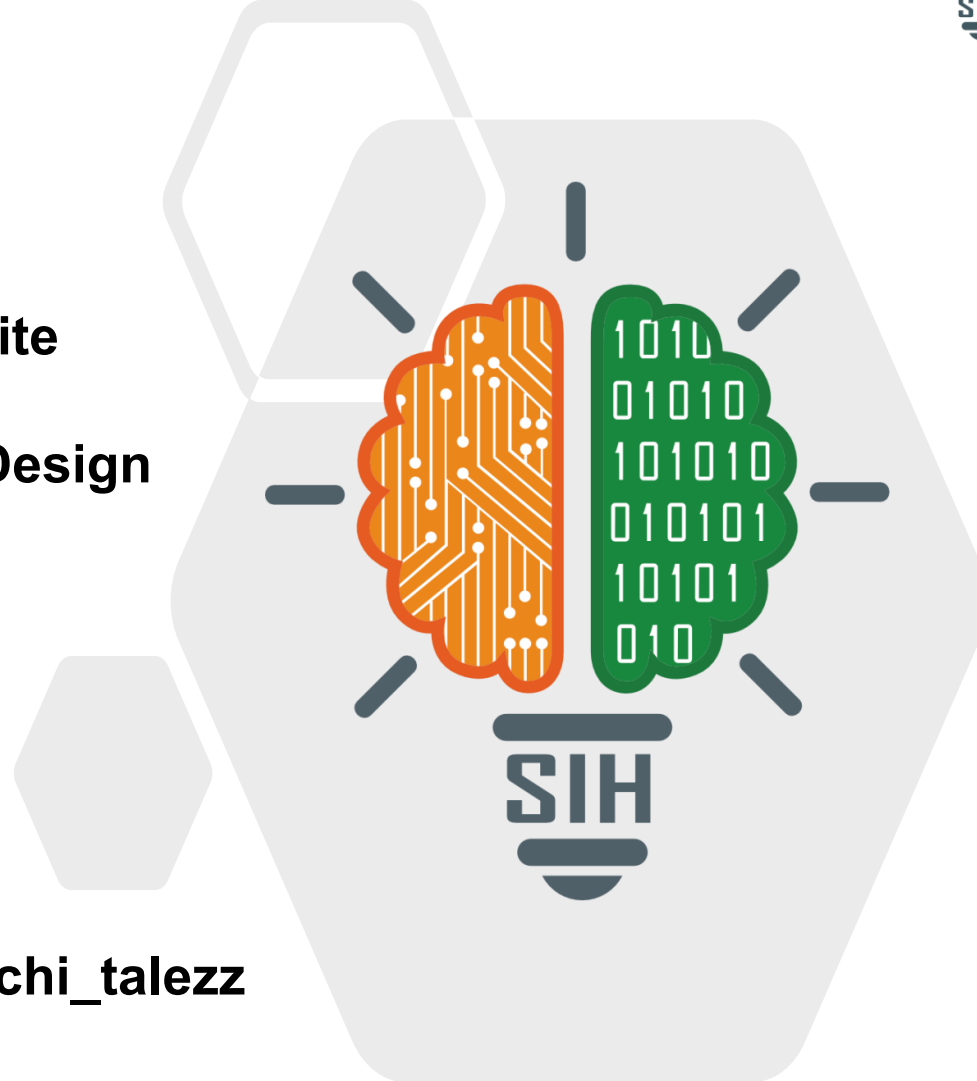


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26114**
- **Problem Statement Title - Smart City Site Planning Using Autodesk Forma Site Design**
- **Theme - Carbon Emission**
- **PS Category - Software/Hardware**
- **Team ID- SIH26-A0H-T505**
- **Team Name (Registered on portal) - Archi_talezz**



SMART CITY PLANNING USING AUTODESK FORMA SITE DESIGN

PROPOSED SOLUTION:

Develop a smart, data-driven site-planning approach using Autodesk Forma to create and compare multiple urban layouts. The alternatives are evaluated for carbon emissions, solar exposure, wind, shading and outdoor thermal comfort to identify the most sustainable configuration. A 3D digital site model and comparison dashboard will visualize the results and support the selection of an optimized, low-carbon and climate-responsive site plan.

Detailed explanation of the proposed solution

Create multiple site-planning scenarios in Autodesk Forma and evaluate their carbon and environmental performance to identify the most sustainable layout.

How it addresses the problem

Enables carbon-conscious decisions at the early planning stage, reducing emissions through optimized building placement, orientation, density and site infrastructure.

Innovation and uniqueness of the solution

Combines 3d Site Planning, Carbon Analysis And Environmental Performance.

TECHNICAL APPROACH



- **Technologies:** Autodesk Forma Site Design, Forma Environmental Analysis, and architectural 3D modelling/visualization tools
- **Methodology:** Site → observe → measure microclimate → map carbon sources → analyse hotspots & emission sources → design intervention → build prototype → test → compare before / after → modify → final output
- **Prototype:** A digital 3D site model demonstrating multiple planning scenarios and their environmental performance, leading to an optimized low-carbon, climate-responsive site plan.

FEASIBILITY AND VIABILITY



Potential Impact on the Target Audience Helps architects, builders and homeowners make low-carbon design choices. Reduces energy consumption and carbon emissions from buildings. Encourages the use of local, recycled and low-carbon materials. Helps users understand the carbon impact of their building decisions. Can be adapted for residential, institutional and commercial buildings. Promotes awareness of sustainable construction practices among students and professionals. Benefits of the Solution. Environmental Reduces embodied and operational carbon emissions. Encourages energy-efficient building design. Reduces construction waste and dependence on high-carbon materials. Promotes locally available and renewable materials. Supports India's long-term net-zero and sustainability goals. Social Creates healthier and more comfortable indoor environments. Increases awareness about sustainable architecture. Encourages communities to adopt environmentally responsible construction. Can support local artisans and traditional construction practices. Economic Reduces energy and maintenance costs over the building's lifetime. Encourages cost-effective local materials and construction methods. Reduces material wastage, lowering construction expenses. Creates opportunities for green construction and sustainable-material industries .One-line PPT conclusion “Our solution aims to make low-carbon buildings easier, more affordable and accessible while creating environmental, social and economic benefits.”

POTENTIAL IMPACT ON TARGET AUDIENCE:

- Enables **architects, urban planners and local authorities** to make data-driven site-planning decisions.
- Helps identify **low-carbon and climate-responsive layouts** before construction.
- Creates healthier and more comfortable outdoor environments for **city residents and users**.

BENEFITS OF THE SOLUTION

ENVIRONMENTAL

- Reduces carbon emissions through optimized site planning and building design.
- Improves microclimate and outdoor thermal comfort through better shading, vegetation and airflow.

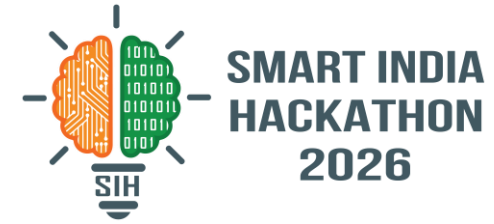
SOCIAL

- Creates more comfortable, healthier and walkable public spaces.
- Improves the quality of outdoor spaces and overall urban living.

ECONOMIC

- Early environmental analysis can help **reduce long-term energy and operational costs**.
- It also minimizes costly design changes by identifying environmental issues **before construction**.

RESEARCH AND REFERENCES



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- [sciencedirect.com](https://www.sciencedirect.com) – Urban Microclimate Review
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